ECON10005 Quantitative Methods 1

Tutorial in week 2: Week 1 Material

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The University of Melbourne

Myself

Zheng Fan

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Don't be shy if you need help

- Join consultation sessions (see time and location on Canvas)
- Visit Ed Discussion Board (make sure you read existing questions before posting)
- Email qm-1@unimelb.edu.au for admin
- Send me an email!

Learning outcome

- Gain experience with which **graphical methods** to use to gauge information from data.
- Gain experience in how to **use Excel to create graphs**.
- Develop skills in **calculating and analyzing changes** in data over time, including absolute and relative measures.
- Be able to **explore trends and shifts** presented in data.

Introduce yourself

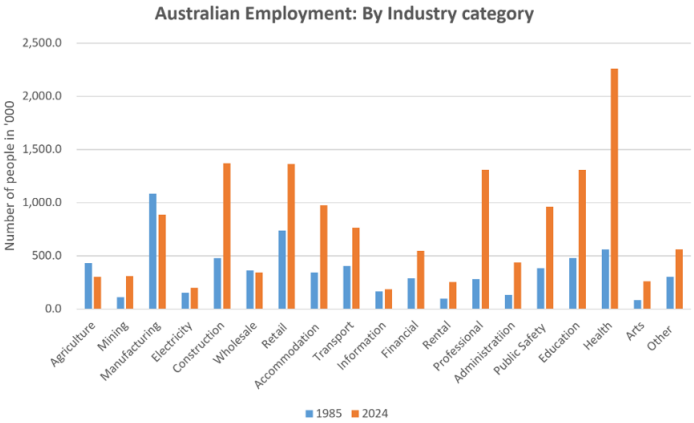
Please take a few minutes to introduce yourself to the people near you.

- Your name
- Your major and area of interest
- One personal interest or hobby

You can also compare and discuss your pre-tute graphs.

Let's target about 5 minutes.

Pre-Tutorial Part A Graphs



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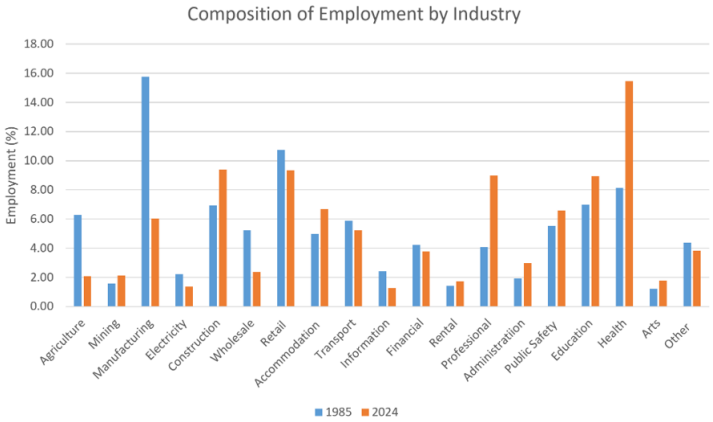

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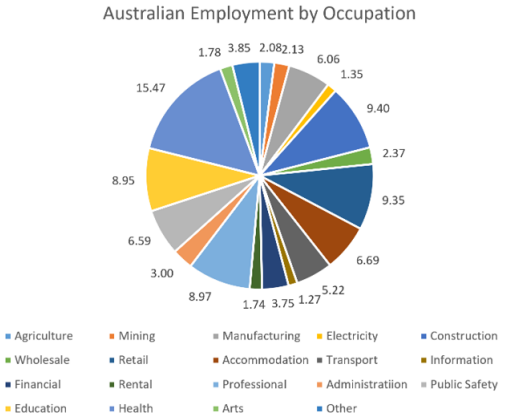

Part A: Tutorial Questions

1. Using the clustered column graphs and summary statistics you created in pre-tutorial exercises for 1985 and 2024, briefly discuss some main features of how the composition of employment has changed in the Australian economy over the last 40 years.
2. Compare the frequency and relative frequency distributions, clustered column graph and pie chart you created for employment data in 2024. Which chart better conveys the composition of employment across industries?

Part A: Tutorial Questions 1

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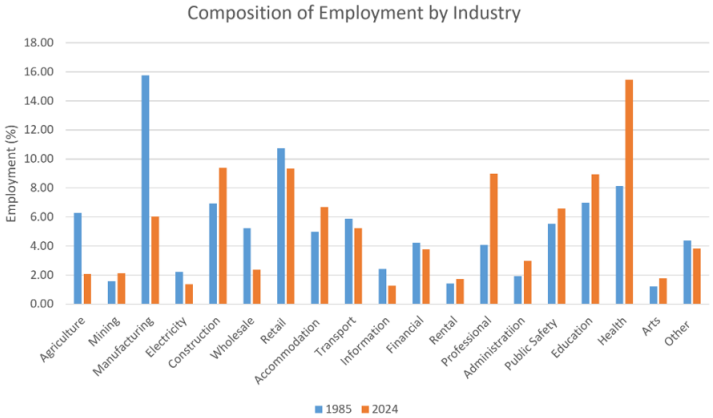

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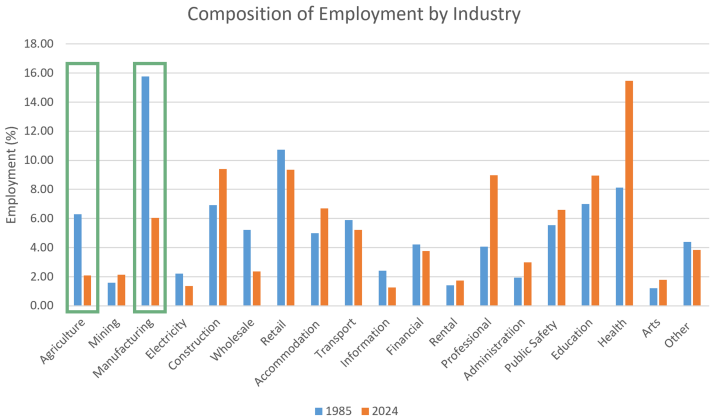

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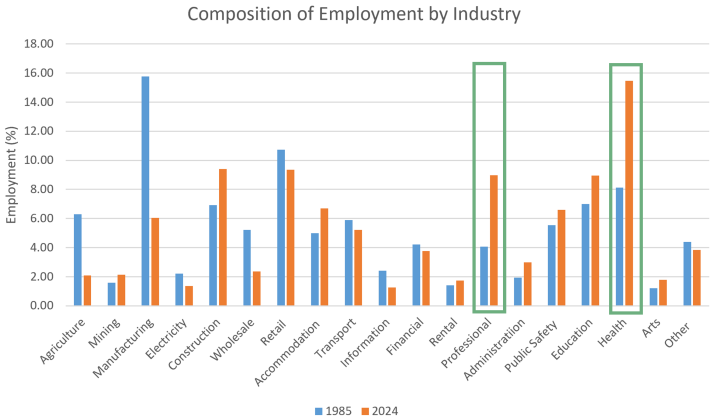

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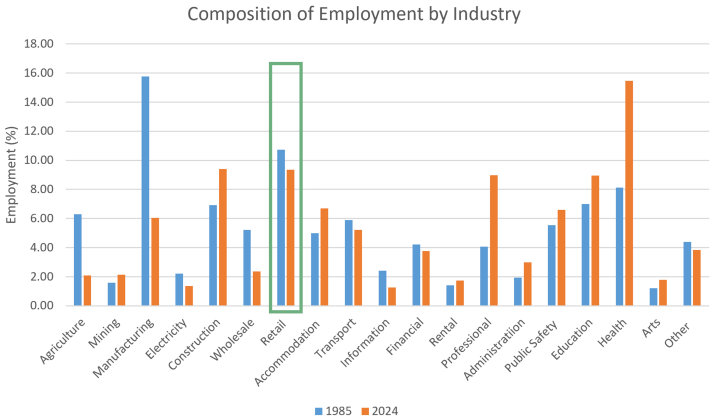


Part A: Tutorial Questions 1

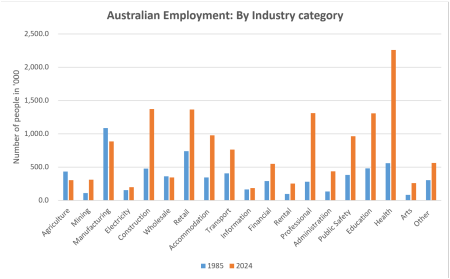
- Australia has transitioned from an economy centered on "growing and making things" into one dominated by service industries.
- There is a shift away from traditional primary and secondary industries.
- The increase in service industries could be driven by demographic changes, such as an aging population and a growing knowledge-based economy.

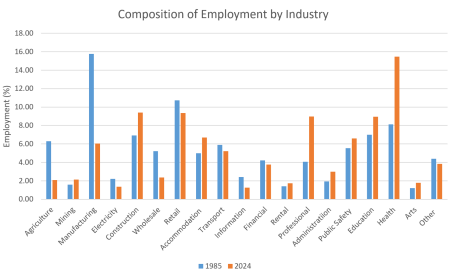
Part A: Tutorial Questions 2

2. Compare the frequency and relative frequency distributions, clustered column graph and pie chart you created for employment data in 2024. Which chart better conveys the composition of employment across industries?

Part A: Tutorial Questions 2

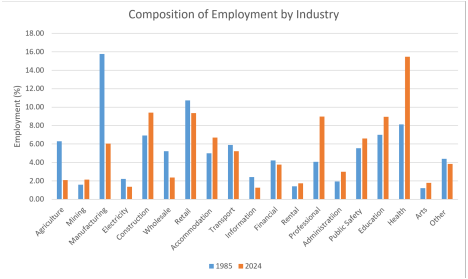
What do each of the below charts tell us?

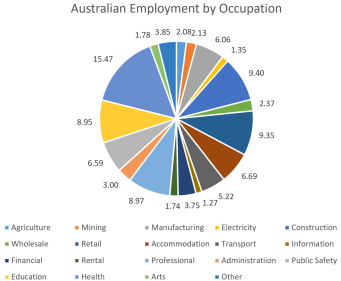




Part A: Tutorial Questions 2

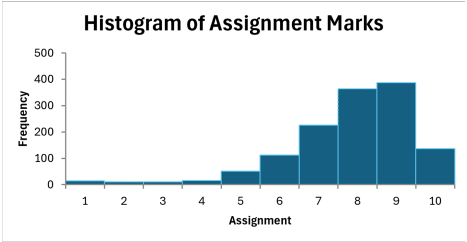
Which chart better conveys the composition of employment across industries?





Pre-Tutorial Part B Graphs

Assignment	Frequency	Relative Frequency (%)
1	14	1.1
2	11	0.8
3	11	0.8
4	16	1.2
5	51	3.8
6	112	8.4
7	226	17.0
8	364	27.4
9	387	29.1
10	137	10.3
	1329	100



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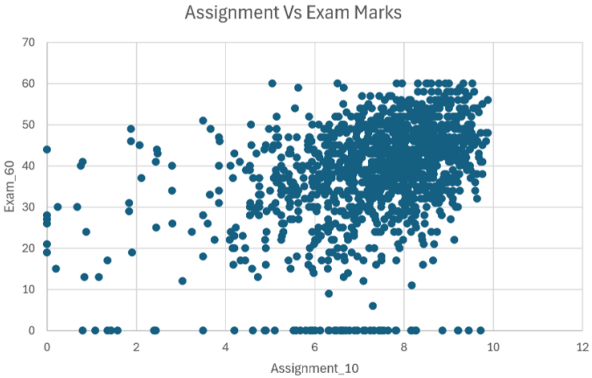

Part B: Tutorial Questions

1. What can you infer about the distribution of marks for each assessment task (e.g., skewness or spread)? Are there any surprising patterns, such as many students achieving very high or low marks?
2. Are there any noticeable clusters, outliers, or patterns in the scatter plots? What might these tell us about the relationship between mid-semester and exam performance?
3. Based on the scatter plots, which mid-semester assessments seem to have the strongest or weakest relationship with exam marks? Discuss why this happens.
4. What does the frequency distribution of grades tell us about overall student performance? Are more students clustered in certain grade ranges (e.g., P or H1), and what might this indicate?

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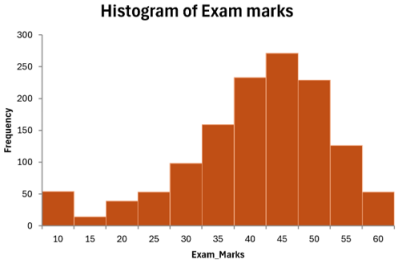

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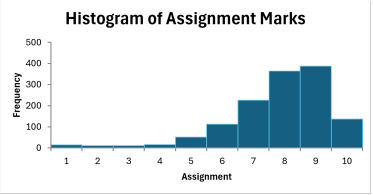
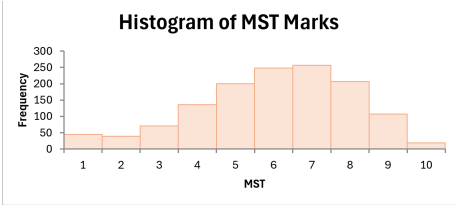
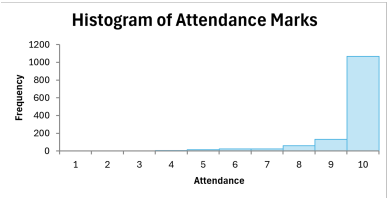
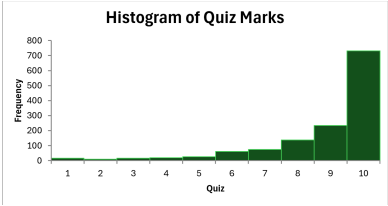




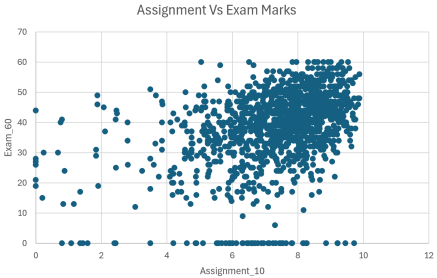


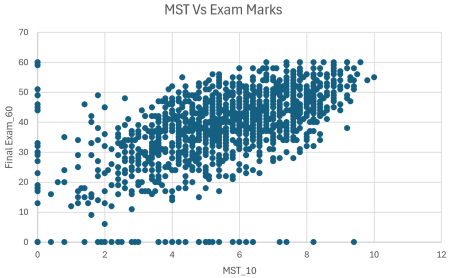


Part B: Tutorial Questions

2. Are there any noticeable clusters, outliers, or patterns in the scatter plots? What might these tell us about the **relationship** between mid-semester and exam performance?

examples:





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3. Based on the scatter plots, which mid-semester assessments seem to have the **strongest** or **weakest** relationship with exam marks? Discuss why this happens.

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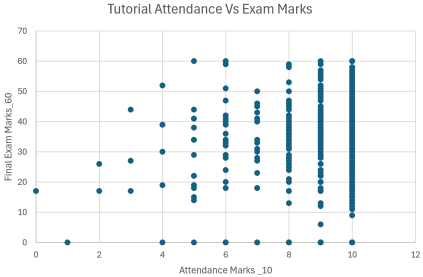

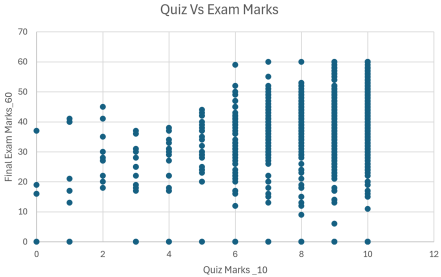

Part B: Tutorial Questions

4. What does the frequency distribution of grades tell us about overall student performance? Are more students clustered in certain grade ranges (e.g., P or H1), and what might this indicate?

Grade	Number	%	Cumulative %
P	151	11.4	11.4
H3	176	13.2	24.6
H2B	190	14.3	38.9
H2A	217	16.3	55.2
H1	362	27.2	82.5
N	138	10.4	92.9
NH	95	7.1	100.0
Total Students	1329	100	

See you next week

Thanks for your attention!

Let me know if you have any questions!